Biochemistry Cume
February 9, 2007

Some Miscellaneous Questions (25 points each)

1. One of nature's dirty tricks is that we are able to convert carbohydrates into fat but not the reverse. Explain why this is true.

2. Although there is no Nobel Prize in Biochemistry as such, many of the recipients in the areas of Chemistry and Medicine and Physiology have been recognized for work on biochemical problems. Describe the work of two Nobel Prize winners in the period 2000-2005 (2006 is explicitly excluded) which fall into this area including their names, where they are presently located or where the work was done, and a short paragraph about the work honored.

3. We used to teach that all enzymes were proteins but are now unable to make such a bold statement. Why? (Be specific in your answer and give examples of non-protein enzyme catalyzed reactions)

4. Other than water, what would you say is the type of molecule present in the largest amount (number of molecules) in a cell? Defend your answer.